

06/15/2025

Numbering Systems - Activity 1

① RESET → ② DRIVE

RESET:

	R	E	S	E	T			
DEC.:	82	69	83	69	84			
1	82 ÷ 2 = 41 R0	69 ÷ 2 = 34.5 R1	83 ÷ 2 = 41.5 R1	69 ÷ 2 =	84			
2	41 ÷ 2 = 20.5 R1	34 ÷ 2 = 17 R0	41 ÷ 2 = 20.5 R1					
3	20 ÷ 2 = 10 R0	17 ÷ 2 = 8.5 R1						
4	10 ÷ 2 = 5 R0	8 ÷ 2 = 4 R0						
5	5 ÷ 2 = 2.5 R1	4 ÷ 2 = 2 R0						
6	2 ÷ 2 = 1 R0	2 ÷ 2 = 1 R0						
7	1 ÷ 2 = .5 R1	1 ÷ 2 = .5 R1						
8	+ 0 → R0	+ 0 → R0						
R	01010010	01000101	S	01010011	E	01000101	T	01010100

DRIVE:

	D	R	I	V	E
DEC.:	68	82	73	86	69
1	68	82	73 ÷ 2 = 36.5 R1	86	69
2			36 ÷ 2 = 18 R0		
3			18 ÷ 2 = 9 R0		
4			9 ÷ 2 = 4.5 R1		
5			4 ÷ 2 = 2 R0		
6			2 ÷ 2 = 1 R0		
7			1 ÷ 2 = .5 R1		
8			+ 0		
D	01000100	R	01010010	I	01001001
R	01010010	E	01000101	S	01010011
V	01010110			E	01000101
T					01010100
XOR:	00010110	00010111	00011010	00010011	00010001
HEX	16	17	1A	13	11

ENCRIPTION MESSAGE: 16171A1311

06/15/25

NUMBERING SYSTEMS ACTIVITY 1

DECRYPTING THE MESSAGE

ENCRYPTED MSG:	16	17	1A	13	11
	00010110	00010111	00011010	00010011	00010001
KEY (DRIVE):	01000100	01010010	01001001	01010110	01000101
XOR :	01010010	01000101	01010010	01000101	01010100
MESSAGE :	D	R	I	V	E ✓